

Homework 5

◇ Frequency response 5.2/5.3/5.4/5.5/5.6/5.8/5.12

5.2 已知某电路的波特图如图 P5.2 所示,试写出 A_u 的表达式。

5.2 Write down the expression of voltage gain A_u according to the Bode Plot in Figure P5.2

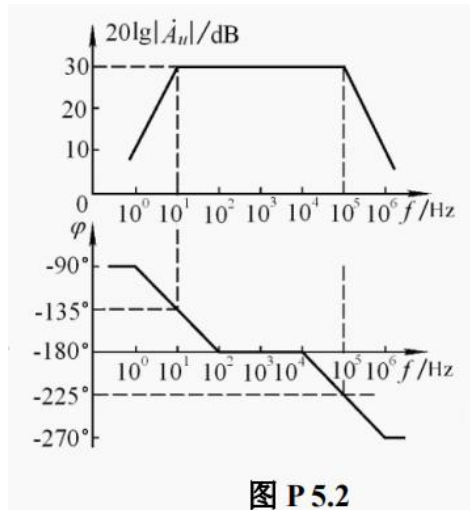


图 P 5.2

5.3 已知某共射放大电路的波特图如图 P5.3 所示,试写出 A_u 的表达式。

5.3 The Bode Plot of a common-emitter amplifier circuit is shown in Figure p5.3. Please write down the expression of the voltage gain A_u .

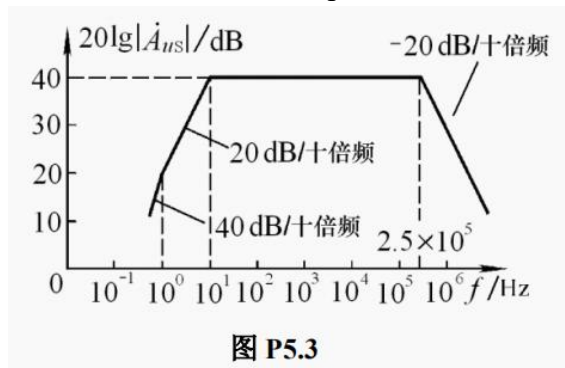


图 P5.3

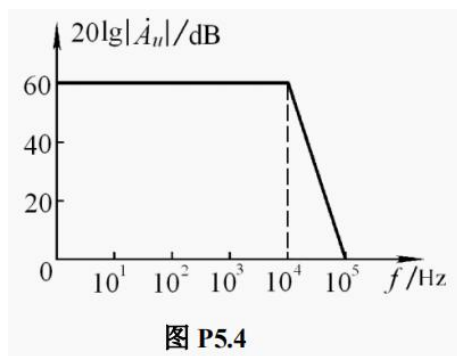
5.4 已知某电路的幅频特性如图 P5.4 所示,试问:

- (1) 该电路的耦合方式;
- (2) 该电路由几级放大电路组成?
- (3) 当 $f=10^4\text{Hz}$ 时,附加相移为多少?当 $f=10^5\text{Hz}$ 时,附加相移又为多少?
- (4) 该电路的上限频率 f_H 约为多少?

5.4 The amplitude frequency characteristics of an amplifier circuit are shown in Figure p5.4, ask:

- (1) What is the coupling mode of the circuit (Direct or RC coupling);
- (2) How many stages are in the amplifier circuit?
- (3) When $f = 10^4\text{ Hz}$, what is the relative phase shift of the voltage gain compared with the midband gain? When $f = 10^5\text{ Hz}$, what is the relative phase shift of the voltage gain compared with the midband gain?

(4) What is the upper cut-off frequency f_H of the circuit?



5.5 已知某电路电压放大倍数

$$\dot{A}_u = \frac{-10jf}{(1+j\frac{f}{10})(1+j\frac{f}{10^5})}$$

试求解 A_{um} , f_L , f_H , 并画出波特图。

5.5 We know the voltage gain of an amplifier circuit:

$$\dot{A}_u = \frac{-10jf}{(1+j\frac{f}{10})(1+j\frac{f}{10^5})}$$

Please (1) get the midband voltage gain A_{um} , lower cut-off frequency f_L , and upper cut-off frequency f_H ; (2) Draw the bode plot of the amplifier circuit.

5.6 已知两级共射放大电路的电压放大倍数

$$\dot{A}_u = \frac{200jf}{(1+j\frac{f}{5})(1+j\frac{f}{10^4})(1+j\frac{f}{2.5 \times 10^5})}$$

试求解 A_{um} , f_L , f_H , 并画出波特图。

5.6 The voltage gain of a two-stage common-emitter amplifier circuit is known:

$$\dot{A}_u = \frac{200jf}{(1+j\frac{f}{5})(1+j\frac{f}{10^4})(1+j\frac{f}{2.5 \times 10^5})}$$

Please (1) get the midband voltage gain A_{um} , lower cut-off frequency f_L , and upper cut-off frequency f_H ; (2) Draw the bode plot of the amplifier circuit.

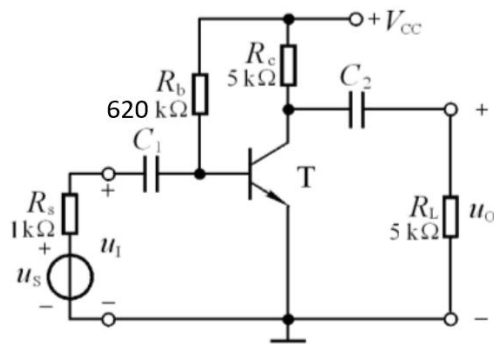
5.8 晶体管交流等效模型利用 h 参数模型（不考虑 π 电容），回路的时间常数为回路的电阻乘以电容。在图 P5.7(b)所示电路中,若要求 C_1 与 C_2 所在回路的时间常数相等,且已知 $r_{be}=1k\Omega$, 则比值 $C_1/C_2=?$ 若 C_1 与 C_2 所在回路的时间常数均为 25ms,则 C_1, C_2 各为多少? 下限频率 $f_L=?$

5.8 Using h-model of a transistor (do not consider the π capacitance in transistor). The time constant of a loop is the resistance times the capacitance in the loop. As shown in the figure below, $r_{be}=1k\Omega$, suppose that the time constant of input loop is equal to the time constant of the output loop, ask:

(1) The ratio $C_1/C_2 = ?$

(2) If the time constant $\tau_{C1} = \tau_{C2} = 25ms$, $C_1=?$ $C_2=?$ and the lower cut-off frequency of the circuit

$f_L = ?$



5.12 已知一个两级放大电路各级电压放大倍数分别为

$$\dot{A}_{u1} = \frac{\dot{U}_{o1}}{\dot{U}_i} = \frac{-25jf}{(1+j\frac{f}{4})(1+j\frac{f}{10^5})}; \quad \dot{A}_{u2} = \frac{\dot{U}_o}{\dot{U}_{i2}} = \frac{-2jf}{(1+j\frac{f}{50})(1+j\frac{f}{10^5})}$$

- (1) 写出该放大电路的电压放大倍数的表达式;
- (2) 求出该电路的下限和上限截止频率 f_L 和 f_H 各约为多少;
- (3) 画出该两级放大电路的波特图.

5.12 We have a two-stage amplification circuit, and the voltage gain of each stage is known:

$$\dot{A}_{u1} = \frac{\dot{U}_{o1}}{\dot{U}_i} = \frac{-25jf}{(1+j\frac{f}{4})(1+j\frac{f}{10^5})}; \quad \dot{A}_{u2} = \frac{\dot{U}_o}{\dot{U}_{i2}} = \frac{-2jf}{(1+j\frac{f}{50})(1+j\frac{f}{10^5})}$$

- (1) Write the expression of the voltage gain the amplifier;
- (2) Find out the lower and upper cut-off frequencies f_L and f_H of the two-stage circuit;
- (3) Draw the Bode plot of the two-stage amplifier circuit.